WEEK 07 • IS321 ENTERPRISE ARCHITECTURE

Models of Organizations & Enterprises

Why People, Culture, and Structure Are as Central to EA as Technology

People & Culture

Organizational Models

Change Management

TODAY'S ROADMAP

What We'll Cover

01	The People Side of EA	Why architecture is as much about people and culture as technology and process
02	Influences on the EA Field	The disciplines — from sociology to computer science — that shaped EA practice
03	The Leavitt Diamond Model	Structure, Technology, People, and Tasks — and how a change in one ripples to the rest
04	The Parsons/Thompson Model	Institutional, Managerial, and Technical levels, and its EA³ Cube connection
05	The Organizational Network Model	How flatter, networked organizations relate teams to Lines of Business
06	Organizations vs. Enterprises	Vertical, horizontal, and extended components, and the segmented approach
07	Culture & Managing Change	Why EA programs succeed or fail on stakeholder buy-in, not just architecture

THE HUMAN SIDE OF ARCHITECTURE

Why People Matter in Enterprise Architecture

> *Enterprise architecture is as much about people and social interaction as it is about processes and resource utilization.*

Understanding each of these aspects is essential to accurate current-state views and meaningful future-state views.

A CONCRETE EXAMPLE

The Home Architecture Analogy

An architect needs to understand the composition, preferences, and activities of the occupants to design an effective home — not just draw walls and wiring.

- How occupants will use each room
- Their day-to-day activity patterns
- Their storage needs

An EA architect needs the same insight into the enterprise's "occupants."

MANAGING THE HUMAN IMPACT

From Current to Future: The People Challenge

Insight into the "people aspect" of enterprises is also essential to developing policy, standards, and an EA Management Plan that will actually be accepted by the enterprise.

↓ to ensure plans are accepted and implemented into workflows

Change Is the Constant

Moving from current to future states of the EA involves changes in processes and how people will communicate.

Change Feels Threatening

Change means moving from what is familiar to something unfamiliar — which is uncomfortable, and sometimes threatening, to many people.

Resistance Should Be Expected

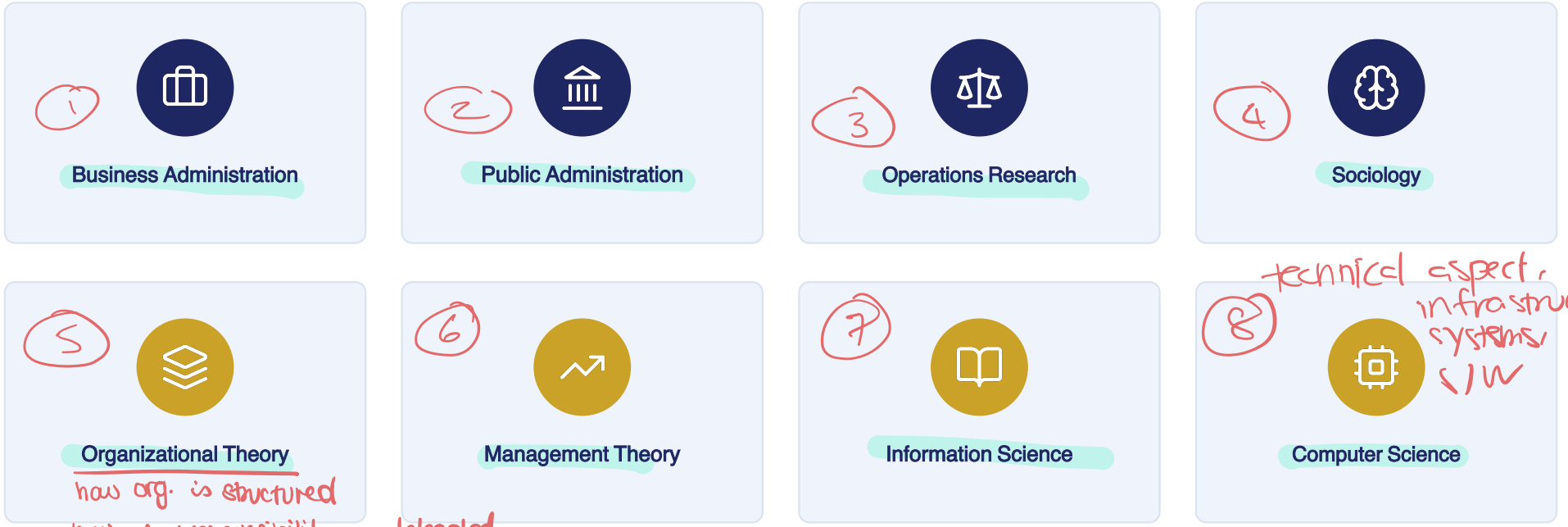
Because of this, there may be resistance to programs such as EA that cause or support changes in resources and processes throughout the enterprise.

→ which stakeholders try to avoid/resist

WHERE EA PRACTICE COMES FROM

Influences on the Field of Enterprise Architecture

Developing an enterprise-wide architecture involves an evaluation and depiction of people, processes, and resources — drawing on theory and practice from many fields:



Understanding the mission, goals, and culture of an enterprise is as important to implementing an EA as the selection of analytic methods and documentation techniques.

A COMMON FOUNDATION

The Structure of Enterprises



The concepts of organizational theory apply directly to enterprises, because enterprises are themselves a type of social organization.

Organizations and enterprises are essentially complex social systems which, regardless of mission, share many similarities in their basic structure and functions.

A CLASSIC MODEL OF ORGANIZATIONAL STRUCTURE

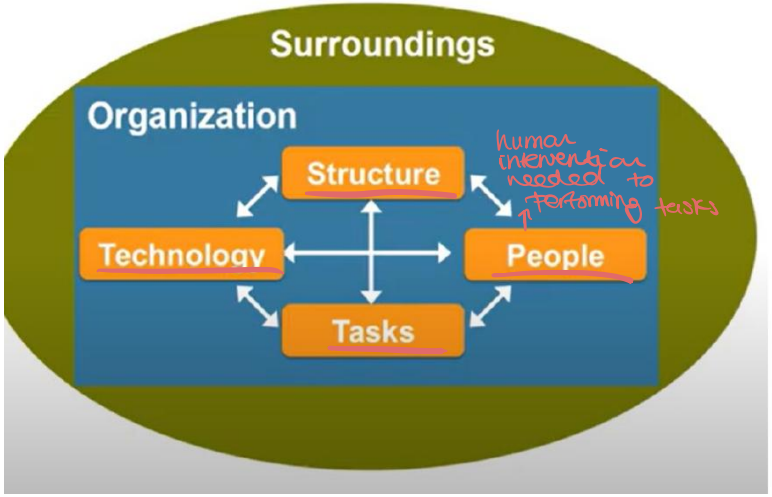
The Leavitt Diamond Model

One of the early models of general organizational structure is the "Leavitt Diamond," presented by Harold J. Leavitt in 1965.

The Core Argument

Leavitt argued that a change in any of the model's four components — Structure, Technology, People, or Tasks — will have an effect on the others, and that the interaction of the components underlies organizational success.

This interdependence is exactly why an EA change in one area — say, new technology — always has ripple effects on people and structure. We'll see worked examples of this shortly.



The Leavitt Diamond: Four Components

Structure

how labor authority, decisions, communication are organized

- Organization of labor
- Decision-making structure
- Communication

People

internal employees + other ppl associated w/ organization
e.g. doctors, nurses, external vendors

- Employees
- People associated with the organization — e.g. external vendors

Technology

tools, facilities, resources supporting day to day operations

→ CRM

- Information technology — IT
- Buildings
- Refurbishment of premises
- Machines
- Hardware and software

Tasks

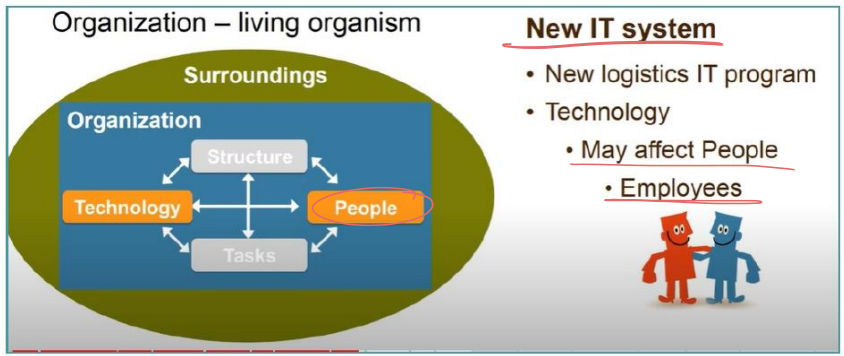
work involved to deliver goods & services

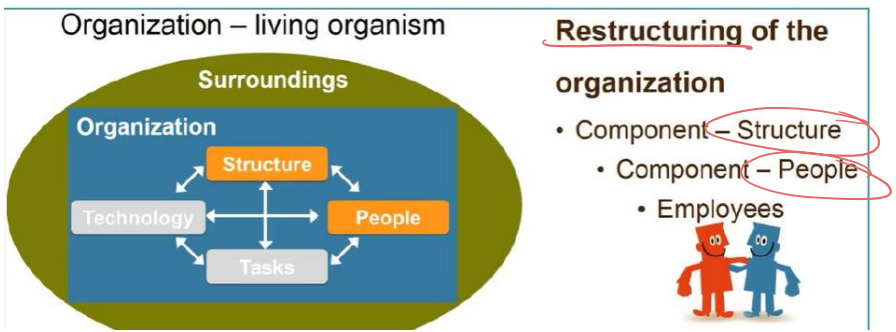
- Goods and services that an organization sells
- For public organizations, the goods and services delivered to the people

The Leavitt Diamond in Action (1 of 2)

A change entering through one component of the diamond ripples through the others — the highlighted boxes show which components are activated.

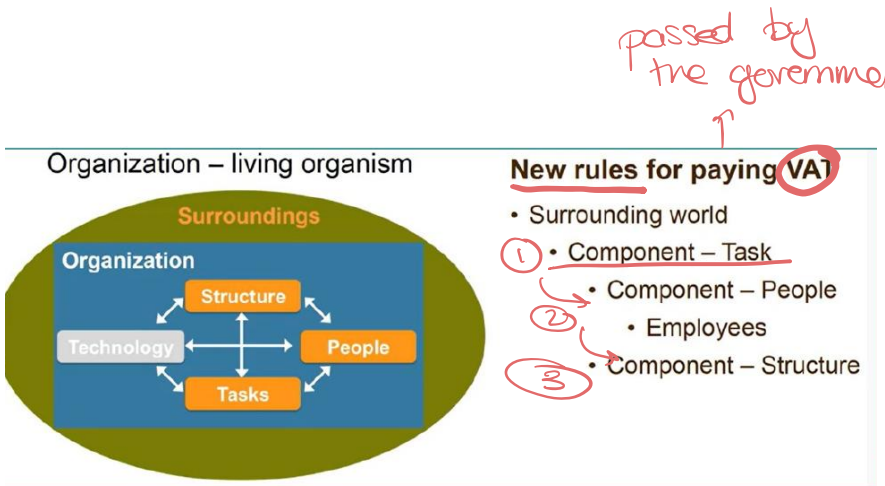
comes in the form of a case study

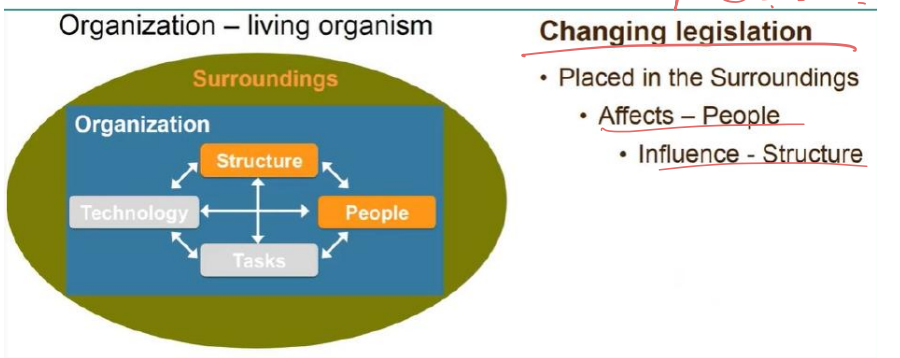




The Leavitt Diamond in Action (2 of 2)

Two more scenarios — an external regulatory change, and a new legislative requirement — activating different combinations of components.

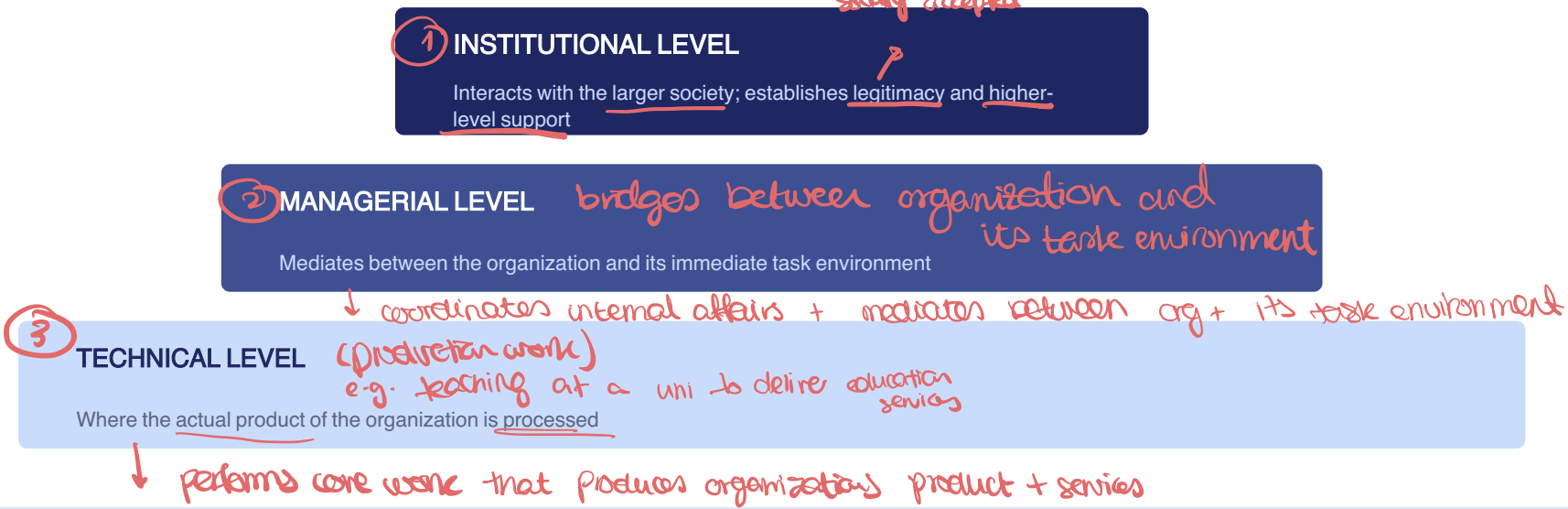




A THREE-LEVEL VIEW

The Parsons/Thompson Model

Another model of general organizational structure is a three-level view, originally envisioned by sociologist Talcott Parsons in the 1950s and further developed by sociologist James Thompson in the 1960s.



Parsons identified three general levels common to most social organizations, based on the observation that different types of activities occur at each level. Thompson built on this by identifying exactly what those activities are — see the table on the next slide.

Purpose and Activities by Level

Organizational Level	*Structure (Parsons' Purpose of Each Level)*	*Function (Thompson's Activities of the Level)*
Institutional	Where the organization establishes rules and relates to the larger society as it derives legitimization, meaning, and higher-level support, thus making possible the implementation of organizational goals.	The organization is very open to the environment in order to determine its domain, establish boundaries, and secure legitimacy.
Managerial	Where mediation between the organization and the immediate task environment occurs, where the organization's internal affairs are administered, and where the organization's products are consumed and resources supplied.	A dynamic of mediation occurs where less formalized and more political activities occur.
Technical	Where the actual "product" of an organization is processed.	The organization is "rational" as it carries on production (input/output) functions and tries to seal off those functions from the outside to protect them as much as possible.

A MORE USEFUL GEOMETRY

From Layers to Circles: The Spherical View

Institutional Level

Interacts with the environment — the outermost ring.

Managerial Level

Facilitates internal resources — the middle ring.

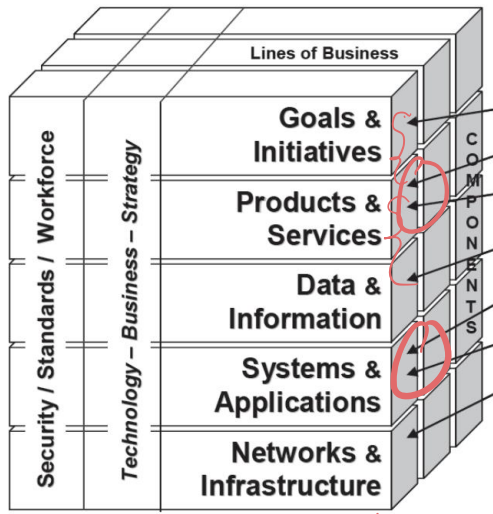
Technical Level

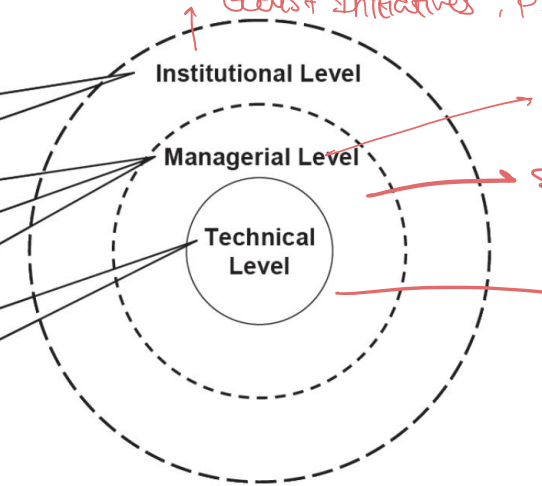
Protects a “core” of essential processes and resources — the innermost circle.

The geometry of the Parsons/Thompson Model has been adapted to resemble a series of concentric circles — a more useful image for depicting a social organization that interacts with its environment via the Institutional Level, facilitates internal resources via the Managerial Level, and protects a “core” of essential processes at the Technical Level. Its value lies in its use as an authoritative reference for developing EA views of structure and process for an organization.

The EA³ Cube & the Spherical Parsons/Thompson Model

Figure 2-4 shows the spherical version of the Parsons/Thompson Model alongside Bernard's EA³ Cube framework — relating the model directly to how an EA framework documents organizational functions.





A FIFTY-YEAR-OLD VIEW, TESTED

Is the Parsons/Thompson Model Still Relevant?

The Question

Regardless of the model's wide acceptance in academia — would this fifty-year-old view still be relevant and useful for understanding the structure of current public and private sector organizations?

Its continued value is its use as an authoritative reference for developing EA views of structure and process for an organization.

THE ANSWER: YES — EVIDENCE FROM PRACTICE

- Many large and medium-sized corporations and government agencies continue to be hierarchical, rule-based, and goal-oriented — primary characteristics of the "rational" organization Parsons and Thompson studied.
- This is still visible in the rational nature of organizational charts, mission statements, strategic plans, operational plans, and business services of these types of organizations.

New Organizations Beyond the P&T Model

New types of organizations have emerged from technology-based changes in how people communicate and work. Global telecommunications and the Internet have made location a largely irrelevant factor for some types of work (e.g., knowledge work and online services), resulting in two primary changes:

1. Regional or Global Organizations

More organizations are becoming regional or global in nature, relying on remote sub-groups to do significant amounts of the work.

2. Self-Employed Knowledge Workers

More people are becoming self-employed knowledge workers who contract their services remotely to various enterprises depending on interest, skills, and availability.

EXAMPLES

People who process digitized health care forms, software developers, web site developers, distance learning instructors, financial traders, insurance salespeople, and telemarketers. Because these organizations can get certain functions accomplished remotely, their structure may become less hierarchical and more collaborative.

The Organizational Network Model (ONM)

New types of organizations and enterprises are appearing which are based on cooperative networks of local and remote individual workers and semi-autonomous teams who carry out key functions.

Greater Efficiency & Flexibility

Achieved by removing layers of management that are not needed in a decentralized operating mode.

Teams Still Reflect P&T Structure

These teams are sub-groups with their own management level and technical level with core processes, so they still exhibit some characteristics of the Parsons/Thompson Model.

The Real Difference

The organization/enterprise's structure is based on these teams and remote workers, whose goals and functions may change depending on internal and external influences.

ONM: Roles and Responsibilities

THE EXECUTIVE TEAM

Sets policy and goals, approves resources, and evaluates results.

Provides to the teams below:

- Policy
- Goals
- General direction

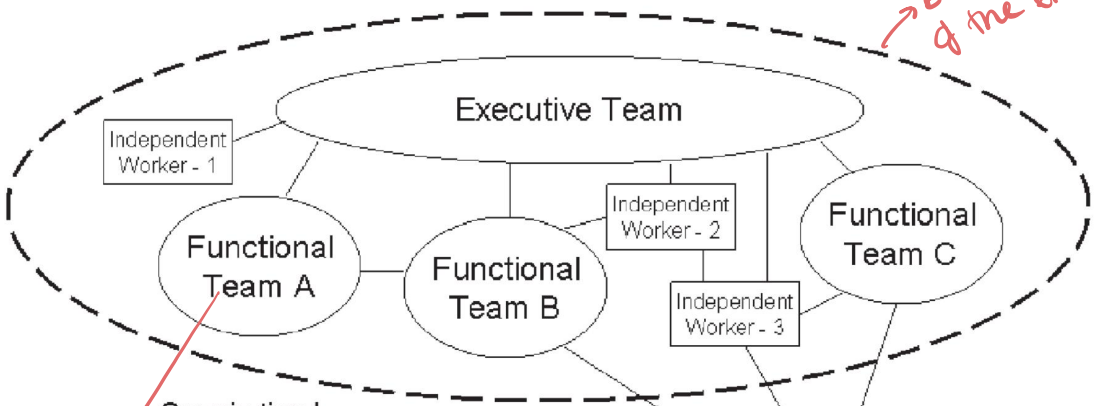
FUNCTIONAL TEAMS & INDEPENDENT WORKERS

Manage ongoing programs and lines of business, new development projects, and team-specific resources.

They receive policy, goals, and general direction from the Executive Team, yet carry out organizational functions in an independent and/or cooperative manner — depending on the goal.

The ONM Diagram

An Executive Team sits at the center of the network. Semi-autonomous Functional Teams and Independent Workers — some inside, some outside the organizational boundary — carry out the work.

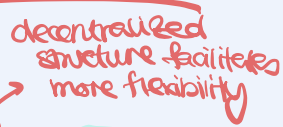


ONM: Built for Flexibility





Responds Faster to Change



Being less hierarchical, these "flatter," more flexible ONM organizations can respond to changing requirements more quickly by creating, modifying, or eliminating Functional Teams and/or adjusting the number and type of Independent Workers.





Extends Beyond One Organization

These ONM organizations and enterprises can also exist as extended supply chains or networks of teams from inside and outside the traditional organizational boundary.





Includes Trusted Partners

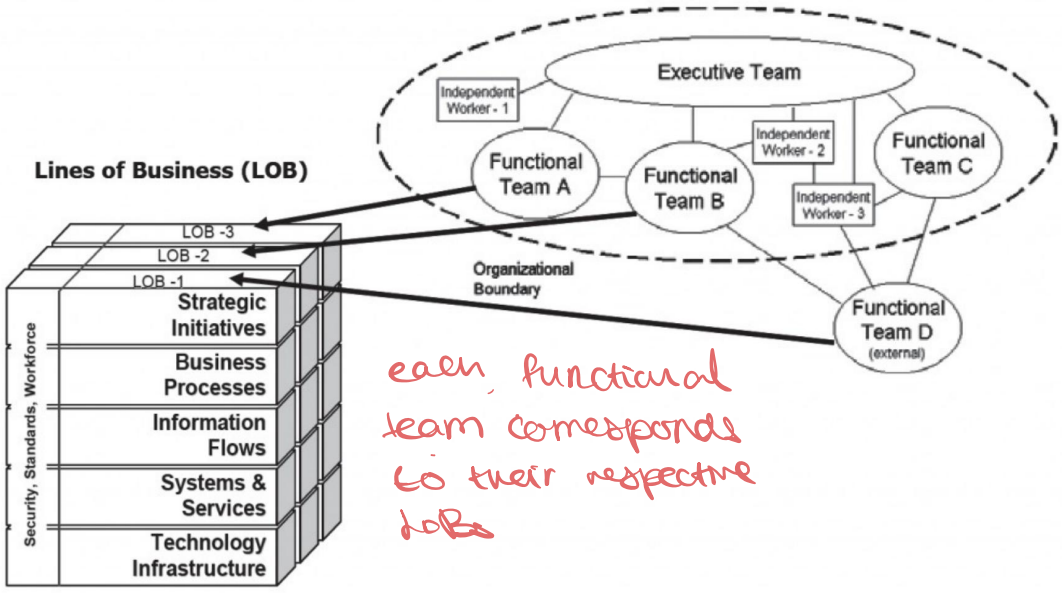
This includes trusted business partners and independent consultants who are allowed to share sensitive information and key resources with the enterprise as part of the activities of the Functional Teams and Independent Workers.

(freelancers)

each functional team manages one LoB

Relating Functional Teams to EA Lines of Business

Functional Teams in ONM organizations can be related to an enterprise's Lines of Business (LOBs) in the EA³ Cube Framework:



TWO RELATED BUT DISTINCT TERMS

Organizations vs. Enterprises

Organizations and enterprises are similar in that both are social entities with a culture, a formal and informal structure, goals, activities, and resources. The difference: an enterprise can be defined as a subset of an organization, or can involve multiple organizations. Enterprises are normally made up of three types of components:

Vertical Components

Also known as lines of business or segments — activity areas particular to one line of business (e.g., research and development).

Horizontal Components

Also known as crosscutting enterprises — more general areas of activity that serve multiple lines of business.

Extended Components

Comprise more than one organization (e.g., extranets and supply chains).

builds one LoB at a time

required equipment needed to conduct surgery

A PRACTICAL WAY TO SCOPE THE WORK

Segments: Building an EA in Pieces

less risky to document a segment

KEY TERM: SEGMENTED APPROACH

EA views of vertical components are complete stand-alone architectures — they contain documentation from all levels of the EA framework. These vertical components are also known as "segments."

1

Document Each Segment

incrementally documents each standalone segment and aggregate it over time

When vertical segments are documented using the same EA framework...

2

Aggregate Into One Picture

...they can be aggregated into a larger architecture picture that may cover several or all lines of business.

3

incremental documentation > large-scale doc incurring more cost

Lower Cost, First Version

This may be preferable for a first EA version — a more manageable amount of work at lower initial cost than attempting the whole enterprise at once.

4

Scales to Decentralized Enterprises

Also useful in large and/or decentralized enterprises, where parts of the architecture may need to be developed and maintained by different groups.

doesn't maintain a single management unit as org may be branched out in

Understanding Culture

> *Understanding the culture of an enterprise is essential to developing realistic views of how strategic goals are established, how processes function, and how resources are used.*

Every Enterprise Is Different

Every enterprise is different in some way — as are its vertical, horizontal, and/or extended sub-enterprises.

Culture Is an Amalgamation

This is because the culture of the enterprise is an amalgamation of the values, beliefs, habits, and preferences of all the people throughout the enterprise or sub-enterprise.

Managing Change

Changes within the enterprise will happen regardless of the presence of an EA program — but without EA, they will happen in a more disjointed or completely independent manner. The effect of the EA program is to coordinate change so that it is driven much more by new strategies and business requirements, and less by new technologies.

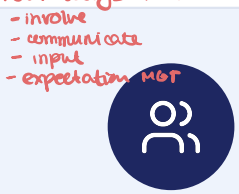
→ prioritizes requirements over emerging tech use

> *To date, major change efforts have helped some organizations adapt significantly to shifting conditions, have improved the competitive standing of others, and have positioned a few for a far better future.*
>
> *But in too many situations the improvements have been disappointing, and the carnage has been appalling — with wasted resources and burned-out, scared, or frustrated employees."*
>
> — John Kotter

explains the impact of introducing change

Why People Resist Change — and How to Help

People can be resistant to changes in their environment, whether at home or in the workplace. If people don't have some level of control, the EA program may be resisted by stakeholders — unless something is done to increase their level of control:



Involve Stakeholders

In the EA program's establishment and management.



Communicate Regularly

Regularly and effectively communicate EA activities to stakeholders.



Allow for Input

Allow for stakeholder input to EA planning and decision-making.



Manage Expectations

Manage stakeholder expectations as to what the EA program can do.

KEY TERM

Change Management

> *The process of setting expectations and involving stakeholders in how a process or activity will be changed, so that the stakeholders have some control over the change and therefore may be more accepting of it.*

Those affected by the EA program are called "EA stakeholders," and they are the ones most likely to resist the program and/or its changes. Stakeholder involvement should reach:

- Governance and oversight activities
- Selection of an EA framework and methodology
- Participation in and reviews of documentation activities
- Development of and updates to the EA Management Plan

Regular, effective communication with all stakeholders is the other half — both formal and informal:

- Formal: EA Program Communication Plan
- Formal: the EA Management Plan itself
- Formal: notices on periodic updates to current/future EA views
- Informal: ongoing communication to sustain participation and support

(day-to-day communication involving follow-ups) e.g through whatsapp

KEY TERM, CONTINUED

“A Place at the Table”

Buy-In Through Governance

The details of EA program governance are discussed later, but it is sufficient to say that it is important to provide “a place at the table” for as many stakeholders as can be accommodated. This increases buy-in for EA policy and decision-making, as well as the success of implementing changes called for in the future architecture.

Expectation Management

Yet another way to promote the success of the EA program and help stakeholders deal with change. It is about identifying realistic outputs and outcomes.

Accomplished by collaboratively assessing: the capability of the EA program to document current and future architectures, the timeframe and resources that will take, and the obstacles to acceptance by stakeholders.

Expectation management is an ongoing aspect of the EA program.

SUMMARY

Key Takeaways

- EA is as much about people and social interaction as it is about processes and resource utilization.
- The Leavitt Diamond shows that Structure, Technology, People, and Tasks are interdependent — a change in one affects the others.
- The Parsons/Thompson Model's three levels (Institutional, Managerial, Technical) remain a valid, authoritative reference for EA views today.
- The Organizational Network Model reflects how flatter, networked organizations relate Functional Teams to EA Lines of Business.
- Enterprises are built from vertical (segment), horizontal (crosscutting), and extended components — enabling a segmented approach to EA.
- Culture and change management — stakeholder involvement, communication, and expectation management — determine whether an EA program succeeds.

Resources & Next Session

Resources

Most slides are directly quoted or paraphrased from the following references:

- Shelly, G. B., & Rosenblatt, H. J. (2011). Systems Analysis and Design (9th ed.)
- Bernard, S. A. (2012). An Introduction to Enterprise Architecture (3rd ed.). AuthorHouse. Kindle Edition.

Next: Week 8

From the people and structures inside an enterprise, we turn to protecting it — Business Continuity Planning, disaster recovery, and why every EA program needs a continuity plan.

Questions?